Xuhui Lu

Ph.D. Candidate | AI for Software Engineering · Reliable Software Systems
Beihang University · Expected Jun. 2027

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Research Interests

- **Software Assurance:** Software specification, program verification and reliable software evolution.
- **Software Agents:** Repository-level software comprehension, automated program repair and reliable AI agents.
- **Software Architecture:** Static analysis, complex-network modeling and deep learning framework analysis.

Education

Ph.D. Candidate, Beihang University *Sep. 2021 - Expected Jun. 2027*

- Advisor: Prof. Zheng Zheng

B.Eng. in Automation, Beihang University *Sep. 2017 - Jun. 2021*

Publications

Journal Articles

1. **X. Lu**, Z. Zheng, F. Qin, X. Ma, and Q. Cai (2025) "Functional and Defect Study in Deep Learning Libraries: A Complex Network Perspective," *IEEE Transactions on Reliability*, 74(4), 4900–4914.
2. X. Wan, Z. Zheng, F. Qin, and **X. Lu** (2024) "Data Complexity: A New Perspective for Analyzing the Difficulty of Defect Prediction Tasks," *ACM Transactions on Software Engineering and Methodology*, 33(6), Article 141, 1–45.
3. W. Ding, Z. Liu, **X. Lu**, X. Du, and Z. Zheng (2025) "KPAMA: A Kubernetes based tool for Mitigating ML system Aging," *Journal of Systems and Software*, 226, 112389.
4. X. Wan, Z. Zheng, F. Qin, **X. Lu**, and K. Qiu (2024) "Adjusted Trust Score: A Novel Approach for Estimating the Trustworthiness of Software Defect Prediction Models," *IEEE Transactions on Reliability*, 73(4), 1877–1891.

Manuscripts Under Review

1. **X. Lu**, Y. Fan, Y. Tang, X. Ma, X. Wan, F. Qin, W. Yuan, and Z. Zheng (2026) "FCG-RAG: Structure-Aware Retrieval-Augmented Generation for Software Comprehension," *ACM Transactions on Software Engineering and Methodology*. Under review.
2. X. Ma, Z. Wang, **X. Lu**, X. Du, C. Li, J. Li, H. Li, and Z. Zheng (2026) "DeepGroundRepair: From API-Grounded Generation to Shape-Aware Repair for Deep Learning Programs," *ACM Transactions on Software Engineering and Methodology*. Under review.

Selected Research Projects

HarmonyOS Application Fault Diagnosis and Repair *2025–2026*

Beihang–Huawei Joint Research Project

- Lead Student Contributor
- Developed an OpenHands-based automated fault localization and iterative repair pipeline for HarmonyOS applications.
- Integrated log and trace analysis, application-logic-driven test generation for fix validation, and patch-impact-guided regression testing to detect newly introduced defects.

Hubei Provincial University–Industry Joint Education Program *2024–2025*

Beihang–Guide Infrared Joint Research Project

- Lead Student Contributor
- Developed software agents to detect inconsistencies between requirements and implementations and localize potential defects.

Software Safety Analysis and Defect Localization *2022–2023*

Collaborative Research Project with Nanjing University and Tianjin University

- Reproduced three real-world defects in deep learning libraries, analyzed their root causes and implemented fixes.

Automated Software Testing and Coverage Analysis *2022–2023*

Collaborative Research Project with a National Research Institute

- Implemented and integrated algorithms for automated test-path generation, test-case generation and coverage calculation.

Fault-Based Reliability Testing and Evaluation for AI Systems *2021–2022*

Beihang–Huawei Joint Research Project

- Compared the architectures of MindSpore, TensorFlow, PyTorch, MXNet, and PaddlePaddle using cross-language function call graphs. Mapped defective functions identified from issue reports onto the call graphs to analyze defect distribution and potential propagation paths.

Research & Industry Experience

Research Engineer, Guide Infrared *Sep. 2023 - Sep. 2025*

- Developed a diffusion-based pipeline for synthetic image generation and image-conditioned video generation to enrich training scenarios. Trained and deployed YOLO-based object detection models on proprietary datasets, improving small-object detection performance through multi-scale feature fusion.
- Developed and evaluated multi-scale template-matching software using the OWL open-vocabulary object detector, and contributed to the development of an automated image annotation pipeline.
- Participated in the research and development of vision-language-action models and world models for humanoid robots.

Research Intern, Huawei *Jul. 2023 - Sep. 2023*

- Analyzed numerical precision issues in MindSpore mixed-precision training on Ascend hardware, including FP16 overflow/underflow and dynamic loss scaling. Investigated software-based fault injection to simulate hardware bit-flip faults in deep learning models and assess their impact on model reliability.

Awards

National Excellence Engineer Scholarship, Beihang University *2023*

First-Class Graduate Academic Scholarship, Beihang University *2022*

Outstanding Student Leader, Beihang University *2022*

Teaching Experience

Teaching Assistant, Beihang University

Fundamentals of Software Technology *2022*

Applied Discrete Mathematics *2021*

Academic Service

Reviewer *IEEE Transactions on Industrial Informatics (TII)*

IEEE International Conference on Software Quality, Reliability and Security (QRS)

Technical Skills

- **Software Engineering:** Static analysis tools, agent harness frameworks, AI infrastructure and RAG system development.
- **Computing Infrastructure:** Three-GPU server assembly, Linux deployment and server administration.
- **Multimodal AI:** Multimodal model training and fine-tuning, distributed training principles and model capability evaluation.
- **Languages:** English (IELTS Overall Band 7.0).